

# Question-generation pipeline

Source image



Scene-graph snippet (input to the generator)

**objects:** {dog (x2), bed, wall, pillow, outlet, sheet, ...}

**attributes:** wall.color = white; pillow.color = blue; bed = large

**relations:** dog on bed; pillow on bed; wall to-the-right-of outlet

*L1/L3 alternate positive and negative phrasings by image index; L4 reuses L2's options unchanged (v8 design).*

## L1 — object presence

"Is there a bed in this image?"

**gold: (A) yes**

## L2 — attribute query

"What color is the wall?"

**gold: (C) white**  
options: {white, +3 distractors}

## L3 — spatial relation (neg.)

"Is the outlet to the right of the wall?"

**gold: (B) no**

## L4 — compositional

"What color is the wall that is to the right of the outlet?"

**gold: (C) white**  
options inherited from L2

## L5 — wordy mirror of L1

"Please read the question carefully and answer it directly. (primary question) The extra wording is only polite framing."

same options, same gold

## L6 — wordy mirror of L2

"Please read the question carefully and answer it directly. (primary question) The extra wording is only polite framing."

same options, same gold

## L7 — wordy mirror of L3

"Please read the question carefully and answer it directly. (primary question) The extra wording is only polite framing."

same options, same gold

## L8 — wordy mirror of L4

"Please read the question carefully and answer it directly. (primary question) The extra wording is only polite framing."

same options, same gold